

Agent-Based Infrastructures for Diachronic Digital Humanities

Cohort

Agents for retrieving, comparing and documenting textual evidence

Shu-Kai Hsieh¹² · Da-Chen Lian¹² · Chunki Lim¹ · Lang-Ching Yeh¹

¹ Graduate Institute of Linguistics, National Taiwan University, Taipei, Taiwan

² Humanistic-AI lab, College of Liberal Arts, National Taiwan University, Taipei, Taiwan

Acknowledgement

Thank you, Michael Radich

Texts

A modified, selected
CBETA-derived corpus

Catalogue

Translator labels and
mixed historical groups

Vocabulary

Curated strings for
Dharmarakṣa research

Cohort builds the profiles and calculates the rankings.

Why we built Cohort

Let researchers direct agents and inspect the evidence behind their proposals.

Today's presentation

Use Buddhist texts to show the functions, where results can mislead, and what the researcher decides.

We are demonstrating research tools, not reporting a new translator attribution.

Agent support for diachronic research

Study texts and their relationships across time.

Corpus retrieval

Find relevant texts and passages.

Semantic analysis

Find passages with related content.

Alignment

Locate matching wording and positions.

Interpretation

Propose explanations and alternatives.

Provenance

Keep sources and checks attached to proposals.

Researcher decisions

Inspect the evidence; accept or reject proposals.

Functions

Corpus

Find passages and read them in context.

Vocabulary comparison

Compare a text's wording with texts labelled by translator.

Inquiry

Give agents a research question and inspect their actions.

Findings

Read the agents' proposals, citations and checks.

Graph

Follow links between sources, proposals and decisions.

Example research questions

Two Lotus translations

Dharmarakṣa 竺法護 · T0263
Kumārajīva 鳩摩羅什 · T0262

Can we find related passages
when the wording differs?

Text and later commentary

Yin chi ru jing 陰持入經 · T0603
Body, mind and senses

Could shared wording mislead us
about who translated this text?

Known relationships let us inspect what the tools get right and wrong.

Find passages and read them in context.

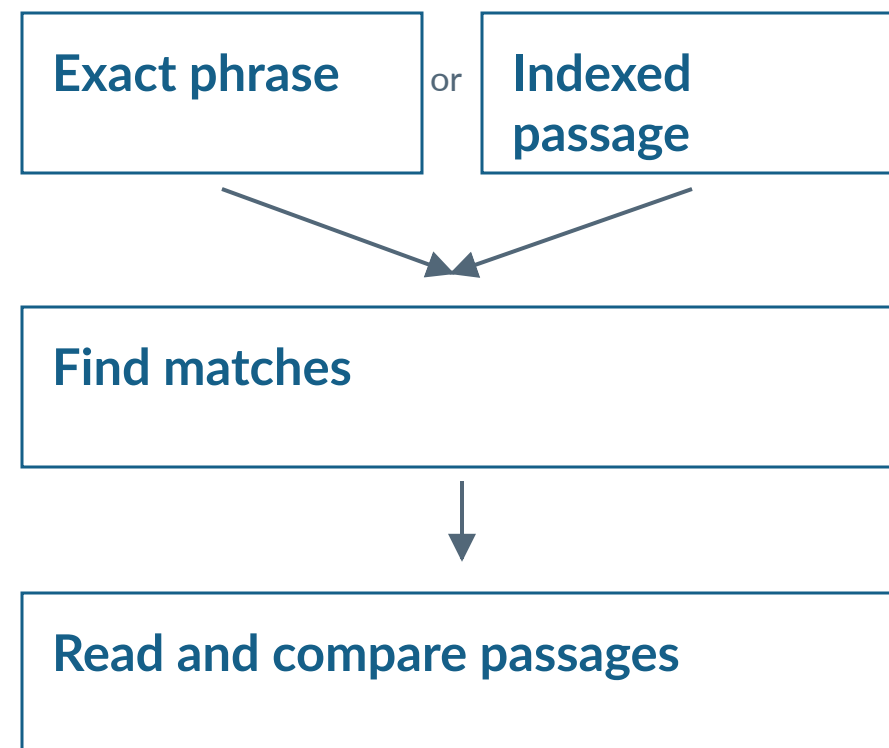
Interface

The screenshot displays the COHORT Corpus interface. At the top, there are tabs for Graph, Findings, Corpus (selected), Vocabulary comparison, and Inquiry. The main section is titled 'Corpus' and contains a search bar with 'Exact phrase' and 'Related passages' options. Below this, a message states: 'Find passages with similar content in other works. Start from an indexed text, not a typed research question.' The 'Embedding model (from filename): mitra-qwen35-embedder' is listed with statistics: '2,160 indexed texts/chapters • 54,616 passages • 400-character windows'. A section titled 'Model and search method' includes an 'Indexed text' dropdown menu showing 'Lotus Sūtra 正法華經 • Dharmarakṣa • catalogue remainder • T0263' and a 'Source position' input field set to '4000'. A 'Find related passages' button is present. Below the search bar, there are tags for 'Lotus Sūtra • Dharmarakṣa', 'Body, mind and senses • T0603', and 'Maitreya • T0453'. The 'SELECTED PASSAGE' section shows 'LOTUS SŪTRA 正法華經 • DHARMARAKṢA • CATALOGUE REMAINDER • T0263-REST' and 'Source characters 4000–4400 (zero-based, end excluded)'. A note indicates '[Source passage hidden in this screenshot]' with 'Previous passage' and 'Next passage' buttons. The 'RELATED PASSAGES' section shows 'Showing 5 of 54,278 candidate passages, ranked by cosine similarity.' The first result is '1. Lotus Sūtra 妙法蓮華經 • Kumārajīva • without Devadatta chapter • T0262-exDevadatta' with a cosine similarity of 0.865 and source characters 2800–3200. A note indicates '[Related passage hidden in this screenshot]' and a 'Check shared wording' button is available. The second result is '2. T0278-ex-earlier-parallels' with a cosine similarity of 0.856 and source characters 208000–208400. A note indicates '[Related passage hidden in this screenshot]'.

Exact search or stored-vector retrieval.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it



Browsing does not add graph records.

How related passages are found

mitra-qwen35-embedder

Selected passage

Use its stored vector

Cosine similarity



Compare with other works

Related passages



Rank candidates to read

Corpus → Related passages; agents can call `semantic_neighbors`.

Model name comes from the index filename; its exact revision is unrecorded.

Example: related content, little shared wording

Earlier Lotus translation

Dharmarakṣa · T0263-rest
Source position 4000

cosine 0.865



Later Lotus translation

Kumārajīva · T0262-exDevadatta
Source position 2800

Longest exact shared sequence: 3 Chinese characters

Shared wording is checked separately, with positions in both sources.

What vocabulary comparison calculates

Vocabulary list

Choose strings to count



Target text

Count those strings



Group profiles

Texts pooled by
catalogue label

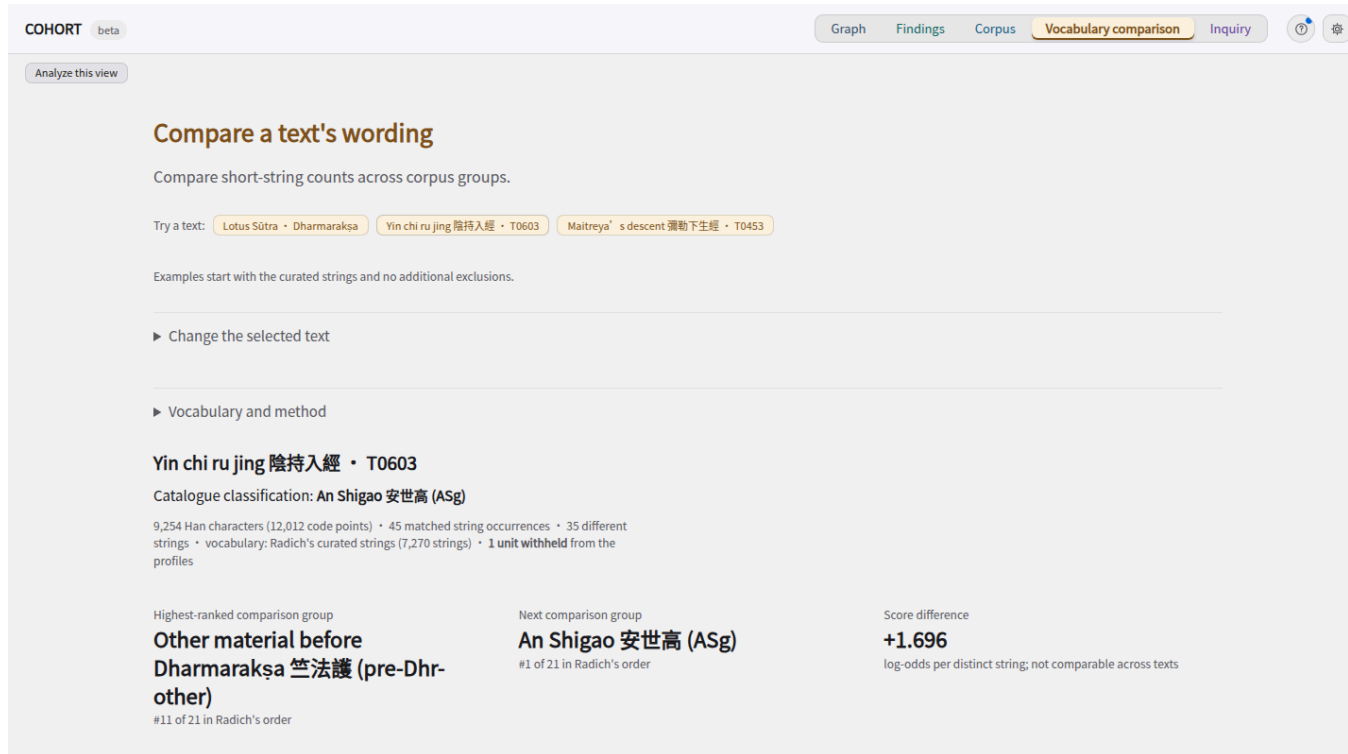


Compare usage → rank the catalogue groups

13 usable groups in this dataset. The target's whole work is withheld.

Compare a text's wording with texts labelled by translator.

Interface



COHORT beta

Graph Findings Corpus **Vocabulary comparison** Inquiry

Analyze this view

Compare a text's wording

Compare short-string counts across corpus groups.

Try a text: Lotus Sūtra • Dharmarakṣa Yin chi ru jing 陰持入經 • T0603 Maitreya's descent 彌勒下生經 • T0453

Examples start with the curated strings and no additional exclusions.

► Change the selected text

► Vocabulary and method

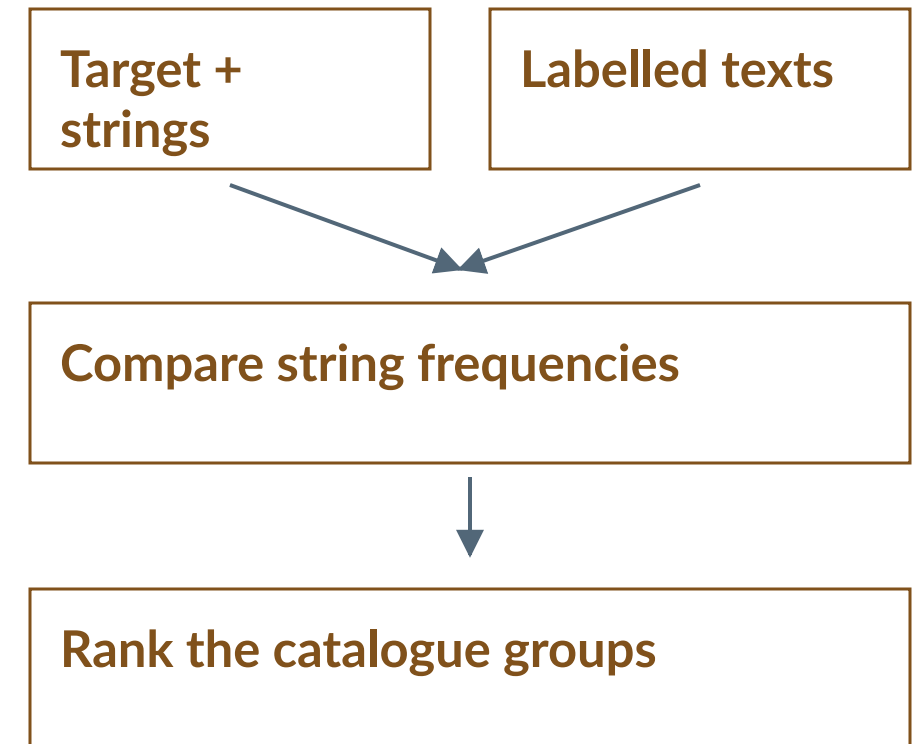
Yin chi ru jing 陰持入經 • T0603

Catalogue classification: An Shigao 安世高 (ASg)

9,254 Han characters (12,012 code points) • 45 matched string occurrences • 35 different strings • vocabulary: Radich's curated strings (7,270 strings) • 1 unit withheld from the profiles

Highest-ranked comparison group	Next comparison group	Score difference
Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) #11 of 21 in Radich's order	An Shigao 安世高 (ASg) #1 of 21 in Radich's order	+1.696 log-odds per distinct string; not comparable across texts

How to use it



Exclude a comparison text and repeat.

The vocabulary selects which strings count.

T0603: remove the later commentary

With T1694

First: Other material before
Dharmarakṣa 竺法護

Margin over next group: 1.696

Without T1694

First: An Shigao 安世高

Margin over next group: 0.096



T1694 is the commentary on T0603. Its wording affects the ranking.

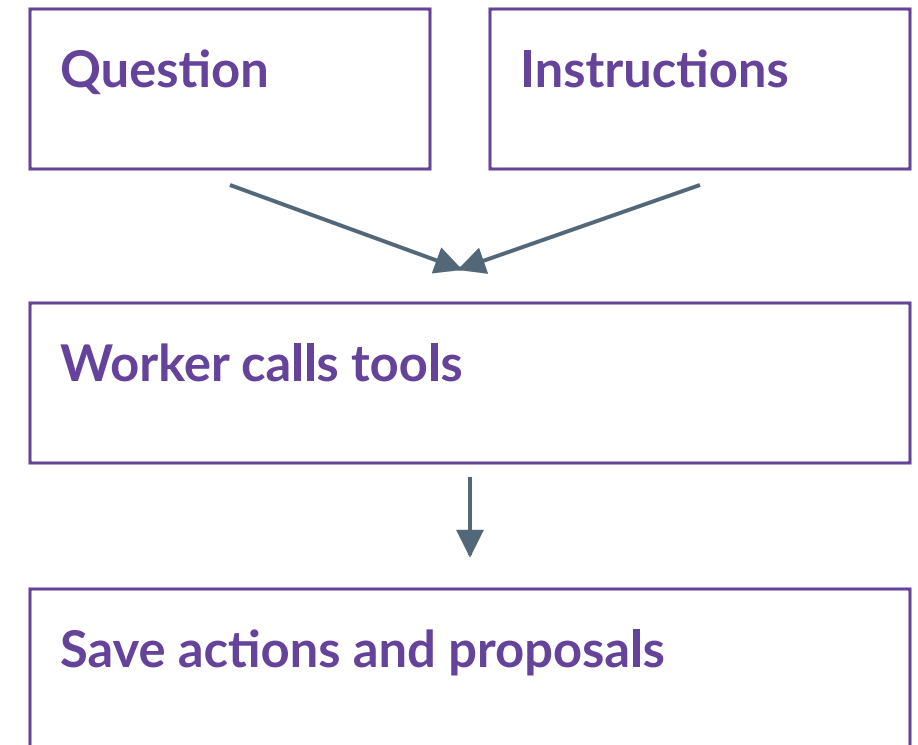
This agrees with the accepted attribution after exclusion; it does not establish one.

Give agents a research question and inspect their actions.

Interface

The screenshot shows the 'COHORT beta' interface with a top navigation bar containing 'Graph', 'Findings', 'Corpus', 'Vocabulary comparison', and 'Inquiry'. The 'Inquiry' panel is active, showing a form for creating a research question. It includes a 'Cancel' button, a 'Choose an example, then edit and record it.' instruction, and three example buttons: 'Find related passages: T0603', 'T0453: find a parallel', and 'Test recurring formulae'. Below these are fields for 'Research question' (containing 'Which passages are related to the Yin chi ru jing (T0603), and what might explain the relationship?') and 'Research instructions' (containing 'Find related passages, compare shared wording, and consider quotation, commentary and recurring formulas. Report source positions and limitations.'). A 'Record question' button is at the bottom of the form. A section for 'Saved questions (4)' with a 'Show hidden questions (3)' button is also visible.

How to use it

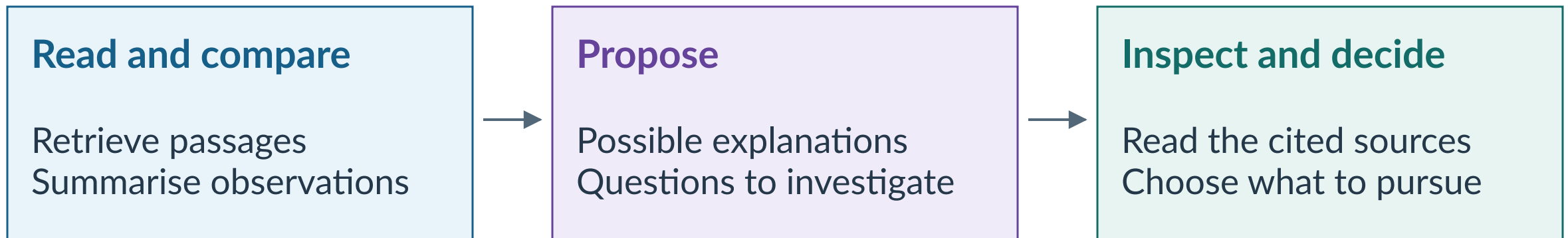


A separate reviewer records checks.

Reopen completed work in Run results.

From passages to research proposals

Agents can compare material, suggest explanations and propose follow-up searches.



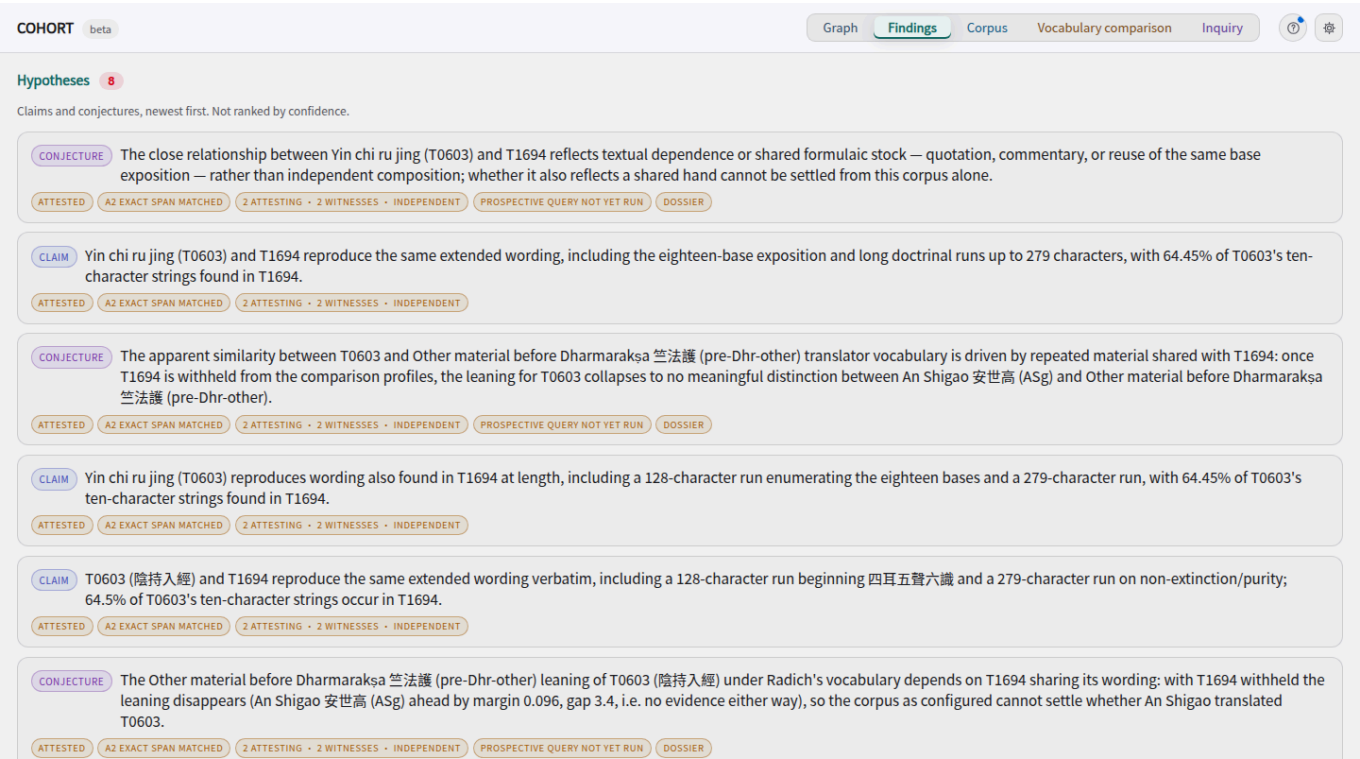
Graph

Keeps proposals linked to passages, checks and researcher decisions.

A citation that resolves can still support a weak interpretation.

Read the agents' proposals, citations and checks.

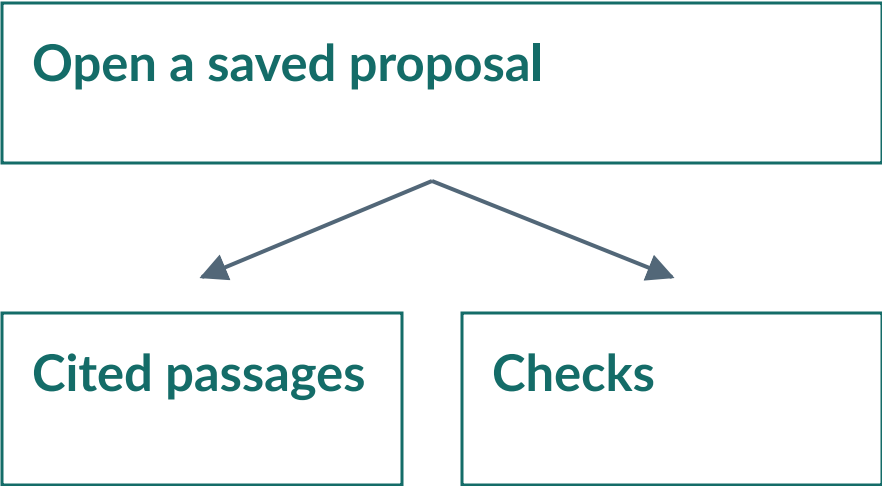
Interface



Follow a citation into Graph to inspect its source.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it

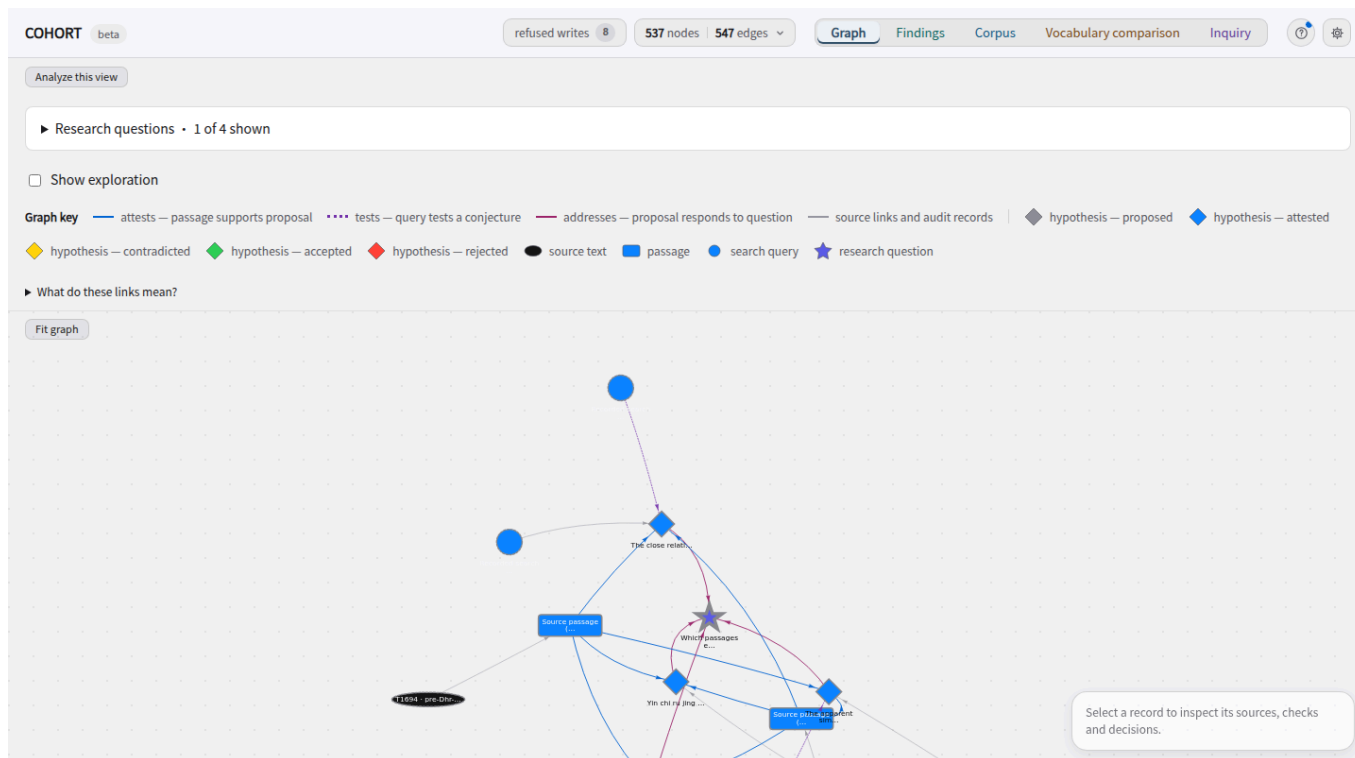


Read its explanation and alternatives.

Inspect one proposal at a time.

Follow links between sources, proposals and decisions.

Interface



Show exploration adds recorded tool activity.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it

Select a research question



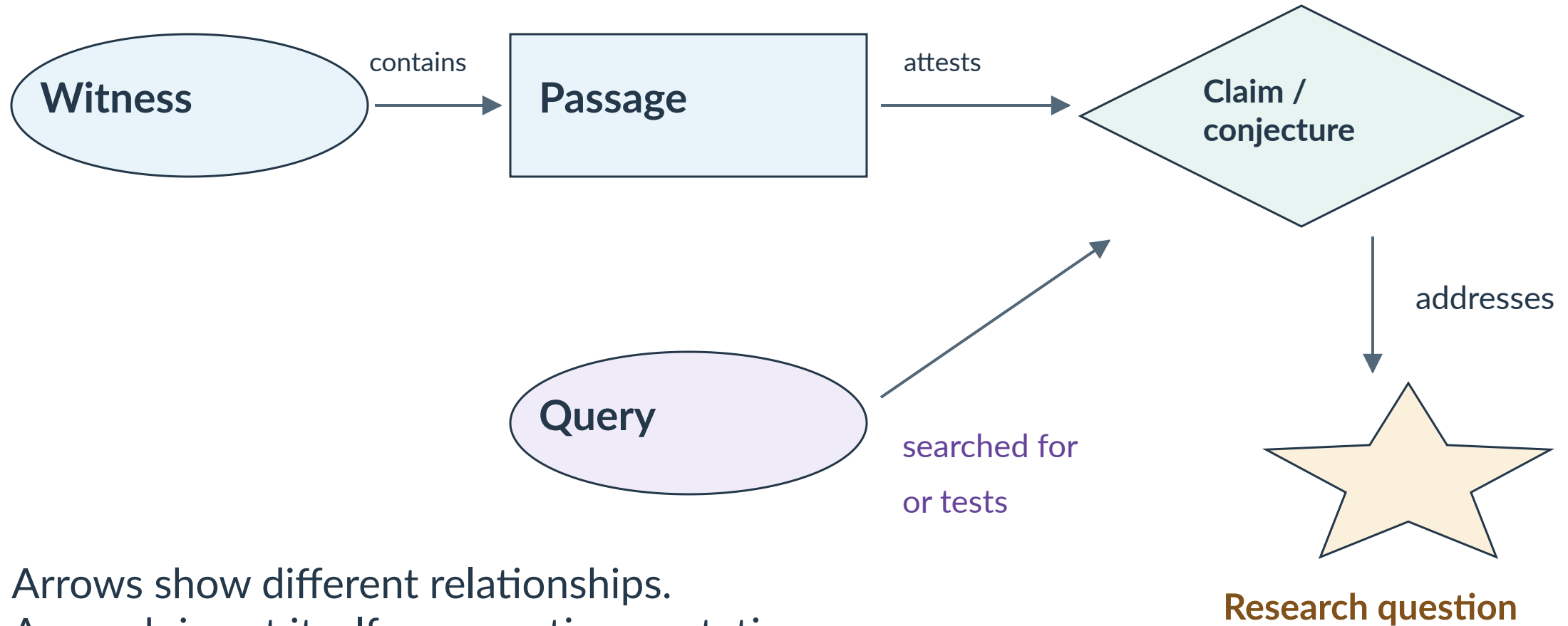
Inspect linked records



Accept or reject a record

Filtering the view does not delete records.

Why a query points to a proposal



Arrows show different relationships.
A search is not itself a supporting quotation.

Checks and researcher decisions

Span verified

The quotation resolves at its source position.

Review recorded

Inspect the reviewer's verdict and evidence.

Researcher decides

Accept or reject an eligible record.

Accepted passage ≠ accepted interpretation

Decisions stay in the record. The source text is unchanged.

Cohort

- 1 Find** Lotus passages and shared wording
- 2 Compare** T0603 with and without its commentary
- 3 Inspect** Saved actions, proposal, citations and checks

Results and limits

Useful outputs

Lotus: a related passage with little exact shared wording.

T0603: exclusion exposes the commentary's effect on the ranking.

Proposals, sources and checks can be inspected together.

Observed limits

Kumārajīva's Lotus wrongly ranks Dharmakṣema first in vocabulary.

One known-answer example cannot validate attribution.

Weak tests and reviewer errors still require researcher scrutiny.